

Nilesh Prasand Pandey

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EDUCATION

University of California San Diego

PhD in Computer Science

Sep 2024- Present

GPA: 3.84/4.0

Research Focus: Inference and Training Efficiency, Model Compression, Hardware-Algorithm Co-design

University of California San Diego

Master of Science in Electrical and Computer Engineering

Sep 2019- June 2021

GPA: 3.84/4.0

Indian Institute of Technology Guwahati

Bachelor of Technology in Engineering Physics

Jul 2015-May 2019

GPA: 8.46/10.0

PUBLICATIONS

Conference Papers

- [1] **Pandey N. P.***, Bhardwaj K.*, Priyadarshi S., Ganapathy V., Esteves R., Kadambi S., Borse S., Whatmough P., Garrepalli R., Van Baalen M., Teague H., Nagel M. "Sparse High Rank Adapters." *NeurIPS*, 2024. (*Equal Contribution)
- [2] Borse S., Kadambi S., **Pandey N. P.**, Bhardwaj K., Ganapathy V., Priyadarshi S., Garrepalli R., Esteves R., Hayat M., Porikli F. "FouRA: Fourier Low Rank Adaptation." *NeurIPS*, 2024.
- [3] **Pandey N. P.**, Kulkarni S., Wang D., Gungor O., Ponzina F., Rosing T. "DPQ-HD: Post-Training Compression for Ultra-Low Power Hyperdimensional Computing." *GLSVLSI*, 2025. **[Oral]**
- [4] Park H., Yessenbayev A., Singhal T., Adhikari N. K., Zhang Y., Borse S. M., Cai H., **Pandey N. P.**, Yin F., Mayer F., Calidas B., Porikli F. "Real-Time, Accurate, and Consistent Video Semantic Segmentation via Unsupervised Adaptation and Cross-Unit Deployment on Mobile Device." *CVPR*, 2022. **[Best Demo Paper Award]**

Under Review

- [5] **Pandey N. P.**, Park J., Gungor O., Ponzina F., Rosing T. "Efficient SLM Edge Inference via Outlier-Aware Quantization and Emergent Memories Co-Design." *In Submission*, 2025.

Workshop Papers

- [6] **Pandey N. P.***, Bhardwaj K.*, Priyadarshi S., Lee K., Ma J., Teague H. "Oh! We Freeze: Improving Quantized Knowledge Distillation via Signal Propagation Analysis for Large Language Models." *ICLR-W*, 2024. (*Equal Contribution)
- [7] **Pandey N. P.**, Nagel M., Van Baalen M., Huang Y., Patel C., Blankevoort T. "A Practical Mixed Precision Algorithm for Post-Training Quantization." *BMVC-W*, 2023. **[Best Paper Award]**
- [8] **Pandey N. P.**, Fournarakis M., Patel C., Nagel M. "Softmax Bias Correction for Quantized Generative Models." *ICCV-W* **[Oral]**

RESEARCH EXPERIENCE

Amazon AGI Labs

Jun - Oct 2025

Efficient RL Training for LLMs

- Proposed Rollout-Aware Adaptive N-Gram Speculative Decoding, introducing a novel intra-prompt and inter-prompt cache storage policy to maximize token reuse across correlated trajectories.
- Demonstrated the method's scalability on the Qwen family (8B-235B), achieving an average $\sim 1.8x$ end-to-end training acceleration and $\sim 2.5x$ rollout speedup across diverse reasoning and interactive agentic benchmarks, validating its utility for large-scale industrial and academic workloads while rigorously preserving training dynamics.
- Manuscript in preparation, open-source contribution to the vLLM library currently in progress.

WORK EXPERIENCE

Qualcomm AI Research

Aug 2021 - Sept 2024

Senior Machine Learning Engineer

Research Contributions

Parameter Efficient Finetuning (PEFT)

- Proposed Sparse High Rank Adapters (SHiRA), a novel adapter achieving up to 2.7% higher average accuracy on language benchmarks than Low Rank Adaptation (LoRA) for LLaMA-7B.
- Achieved 10x faster adapter switching compared to LoRA during inference, demonstrating SHiRA's suitability for edge devices.
- Achieved superior multi-adapter fusion, reducing interference and improving performance across language and vision tasks.

Low Bit Quantization of Generative Models

- Proposed a bias correction technique for INT8 softmax, enabling 40% lower inference latency on mobile devices within W8A16 settings. Achieved 2.7dB SQNR gain and 4.8-point perplexity reduction on Stable Diffusion and OPT-125m.

Development Contributions

- Led quantization efforts in multiple interactive conference demonstrations - Real time on-device segmentation (CVPR'22), Efficient on-device INT4 super resolution (NeurIPS'22), World's fastest controlnet on mobile phone (CVPR'23).

ACHIEVEMENTS

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| 1 | DAC Young Fellow | 2025 |
| 2 | University of California San Diego CSE Department Fellowship | 2024-25 |
| 3 | Publication selected among top 3 papers for Oral Presentation at RCV workshop, ICCV'23 | 2023 |
| 4 | Publication selected for Oral Presentation at CADL workshop, BMVC'23 | 2023 |